

## Assignment 1 (General Materials Chemistry)

Course: Advanced Materials Chemistry

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Max Marks: 50

Q.1. How many types of crystal systems are there and what classifies one crystal system from another? 2

Q. 2. The coordination number for closest packed crystal structure 1

(a) 12 (b) 11 (c) 8 (d) 16

Q.3. What are the Miller indices of a plane that makes intercepts of  $6\text{\AA}$ ,  $3\text{\AA}$  and  $4\text{\AA}$  on the coordinate axes of an orthorhombic lattice with the axial lengths as:  $a = 12\text{\AA}$ ;  $b = 3\text{\AA}$ ;  $c = 8\text{\AA}$

2

Q.4. If in a cubic system, the interplanar spacing of (333) planes is  $2\text{\AA}$ , then  $d$  for (111) planes would be? 2

Q. 5. How many lattice points are there in unit cell of C-centred lattice. What are their co-ordinates? 2

Q.6. Po forms a simple cubic unit cell;  $a = b = c = 3.4\text{\AA}$  and  $\alpha = \gamma = \beta = 90^\circ$ . If density of Po is  $9\text{ g/cc}$ . Then how many unit cells would be there in  $100\text{ g}$  of Po? 3

Q.7. What do you understand by basis? What is the basis of  $\alpha$ -Fe that crystallizes in bcc and NaCl that crystallizes in FCC? 2

Q. 8. How does crystal structure change when ZnO crystallizes as zinc blende and wurtzite? 2

Q. 9. State True or false (Justify): 10

(a) CsCl crystallizes in body-centred lattice.

(b) Co-ordination polyhedral of Na and Cl in rocksalt structure and Ni and As in NiAs structure are same.

(c) The covalent nature of bond leads to high co-ordination no.

(d) Lattice energy does not depend on charges of the ions constituting the compound.

(e) UC and PuC crystallize in rock salt structure.

Q. 10 If lattice energy for  $M_1X$  is 100 kJ/mol, what is your prediction for lattice energy for  $M_2Y$ ; given  $r(M_2) = 3 r(M_1)$  and charge on  $M_1$  is twice that of  $M_2$ ? **3**

Q. 11. What are the two ways in which the perovskite unit cell can be drawn? **3**

Q. 12. What is the difference between defect fluorite structure  $A_{1-x}B_xO_{2-x/2}$  and pyrochlore  $A_2B_2O_7$  (where A is tetravalent and B is trivalent ion). **3**

Q. 13. If an element A crystallises in bcc lattice (a) what is its co-ordination no. (b) its atomic diameter if lattice parameter is "a". **2**

Q.14. Draw the following planes: (110) (210) (220) (421) (432) **5**

Q. 15. Fill in the blanks: **8**

1. A crystal can be defined as the lattice with ..... added to each lattice point.

2. The no. of ZnS formula units in zinc blende structure and rock salt structure are..... and ..... respectively

3. AgCl can be described as cubic close packing of  $Cl^-$  ions with  $Ag^+$  occupying..... Holes.

4. In space filling polyhedral arrangement NaCl can be shown as ..... sharing  $NaCl_6$  octahedra.

5. Ferroelectricity in  $BaTiO_3$  is due to ..... (complete it)

6. Two applications of materials with layered structures are: ..... and .....